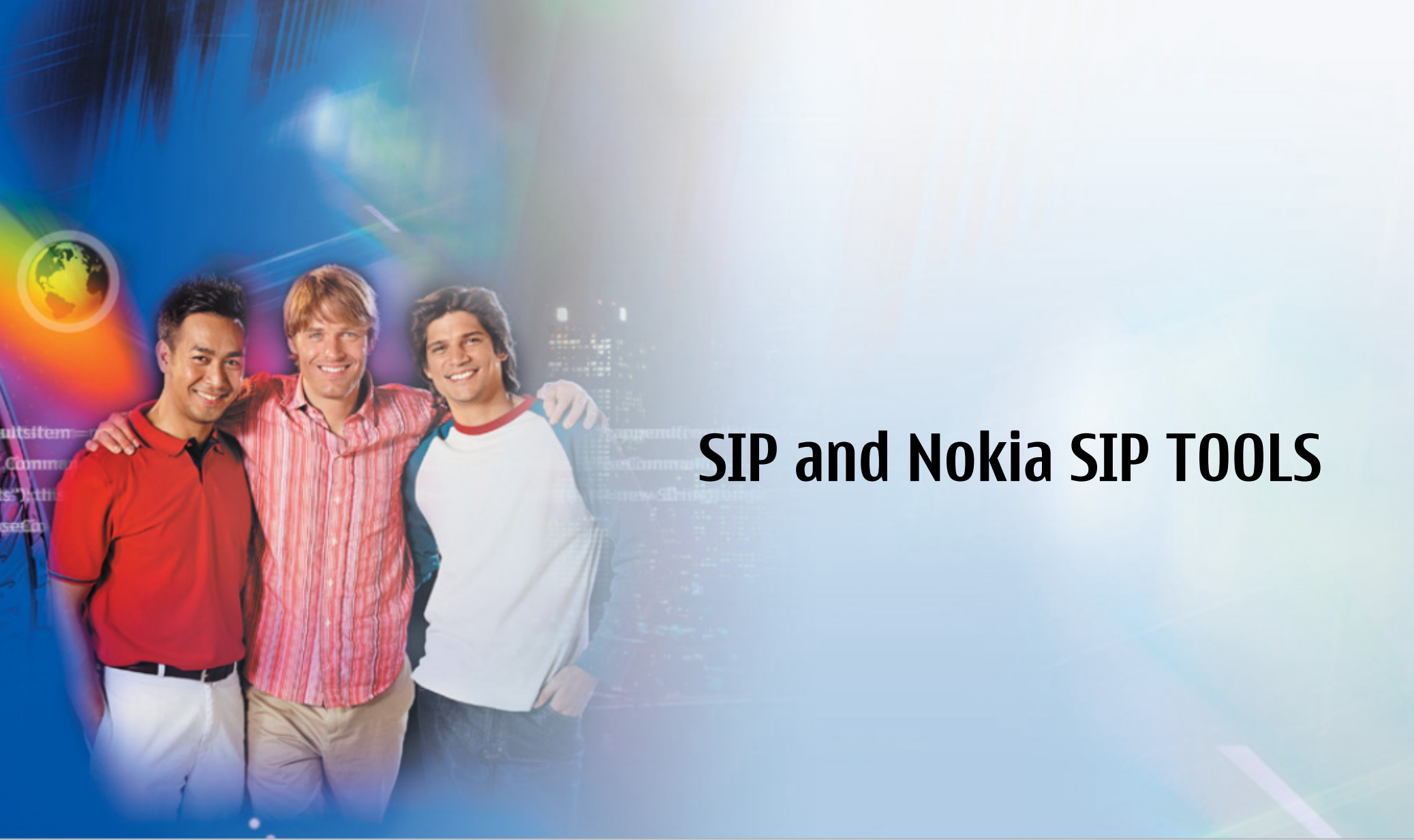


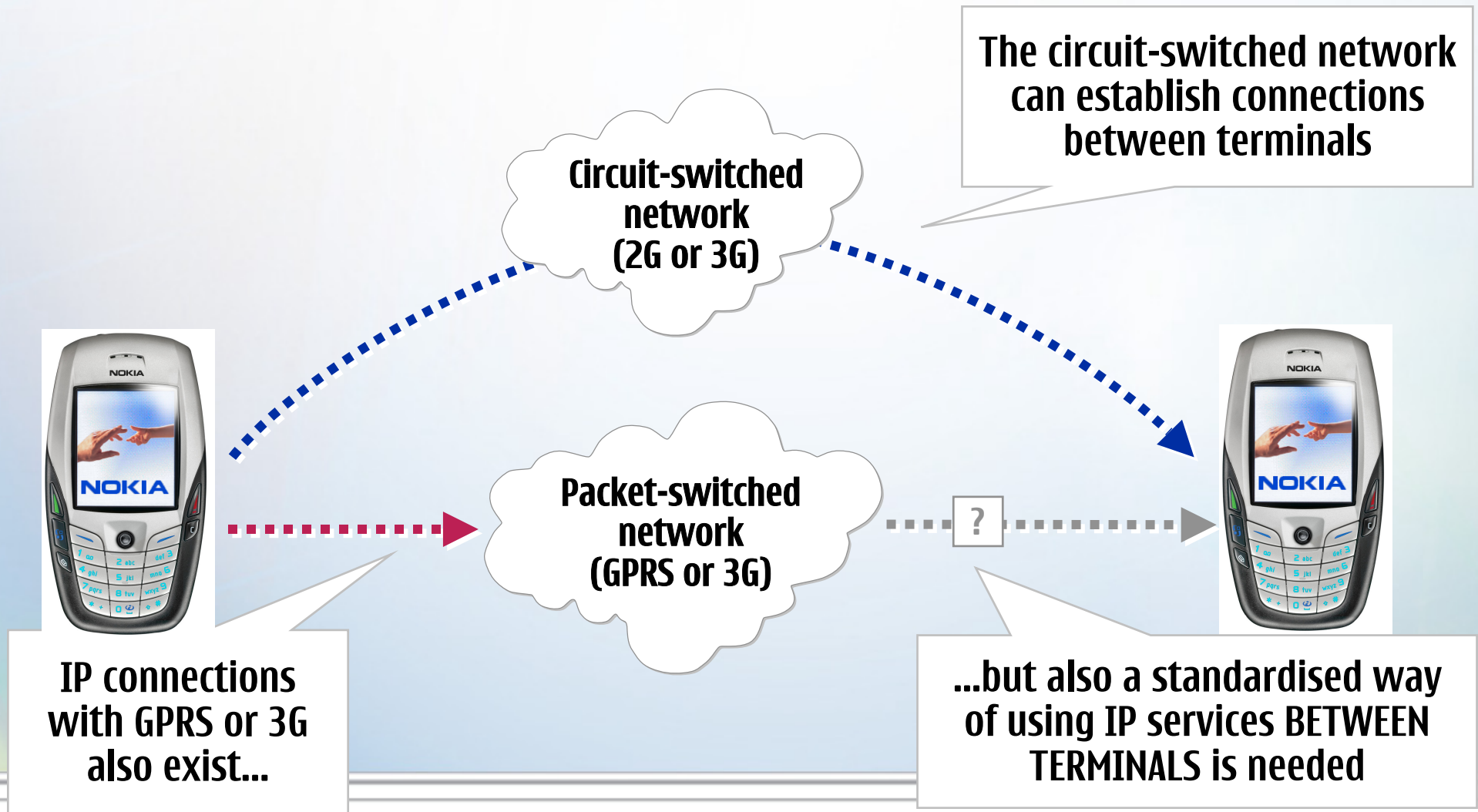


SIP and Nokia SIP TOOLS

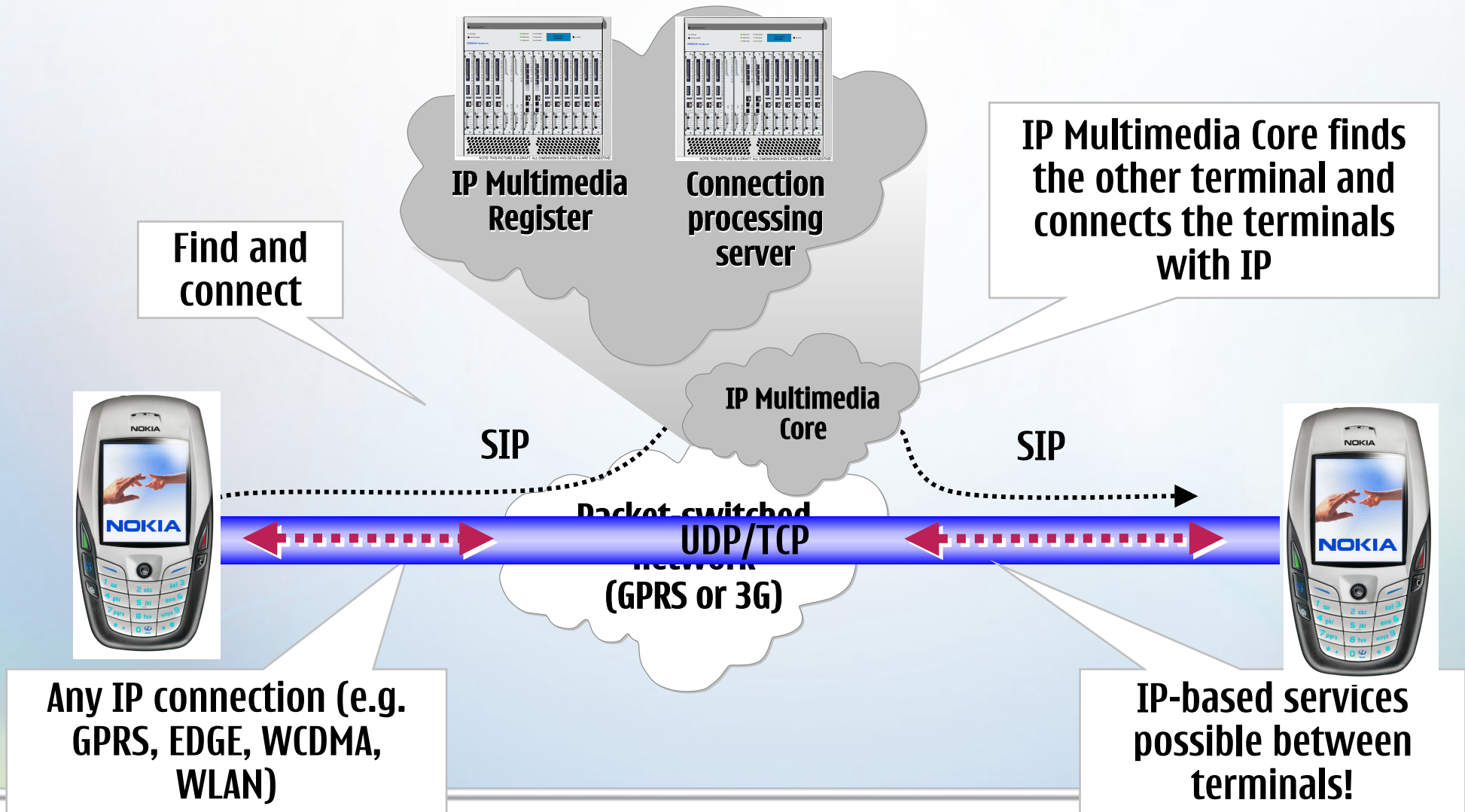
Content

- **Person to person IP connections**
- **What is SIP....**
- **SIP as a service enabler**
- **Nokia Tools and support for SIP**

Connecting people for multimedia?



Person to person IP connection



What is SIP...

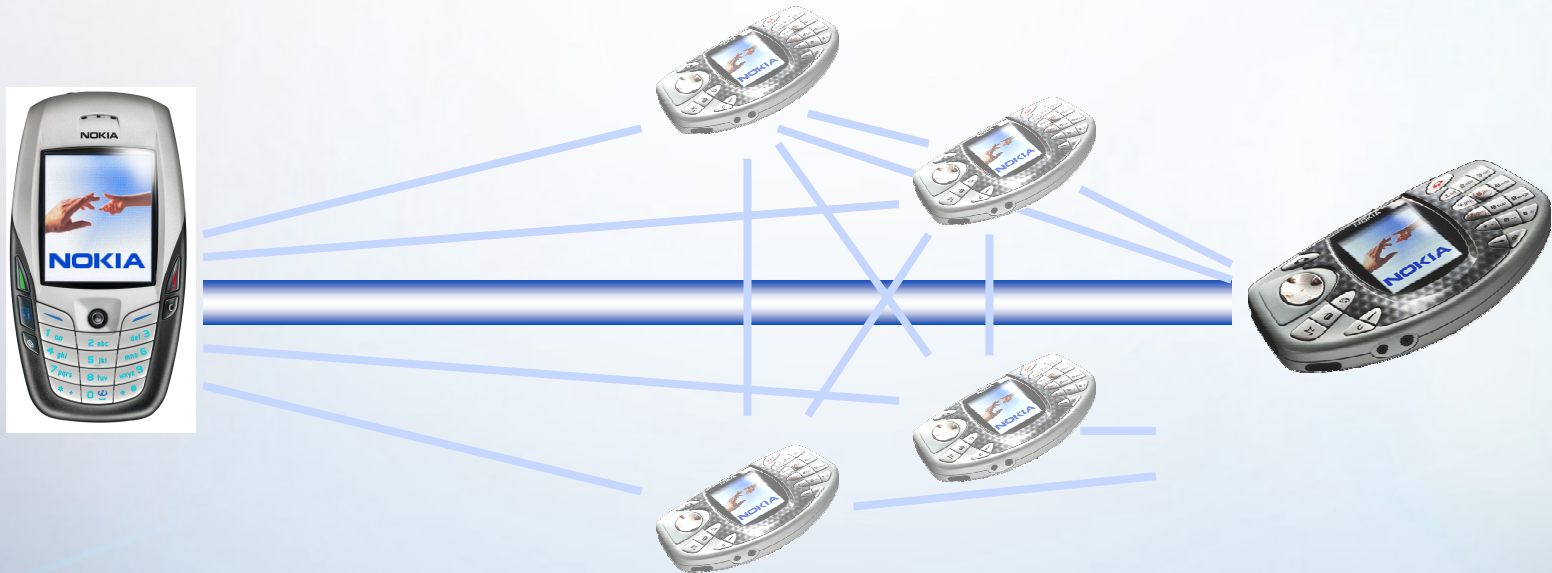
- **Session Initation Protocol**
 - defined by the Internet Engineering Task Force (IETF)
 - designed after HTTP
- SIP is used for initiation of interactive communications sessions between users. SIP also handles termination and modifications of sessions as well.
- Although originally designed for multimedia sessions
 - Scope has been extended: presence, gaming, IM, etc etc
 - SIMPLE: SIP for Instant Messaging and Presence Leveraging Extensions
- Smarter than your "usual transport"
 - personal addressing, routing, etc etc
- Since its approval in early 1999 as an official standard, SIP has gained tremendous market acceptance for signalling communications service on the Internet.
- Adopted by 3GPP in Release 5
 - all Mobile Operators know that they will be operating SIP infra
- More information from IETF RFC3261

Benefits of SIP...

- SIP is a text based language and very similar to HTTP. Easy to program compared to legacy mobile network protocols.
- SIP is gaining acceptance in fixed internet as the protocol for session establishment. Competence developed in the internet community can be utilised in mobile domain.
- Use of IP and SIP enables the convergence of fixed and mobile internet domains.
- SIP can be used to implement all types of peer-to-peer applications...
 - voice
 - video
 - messaging
 - gaming
- ... ultimately replacing existing circuit switched services (voice call, SMS)

SIP as a service enabler

**What we can offer in IP based world
is something different:**

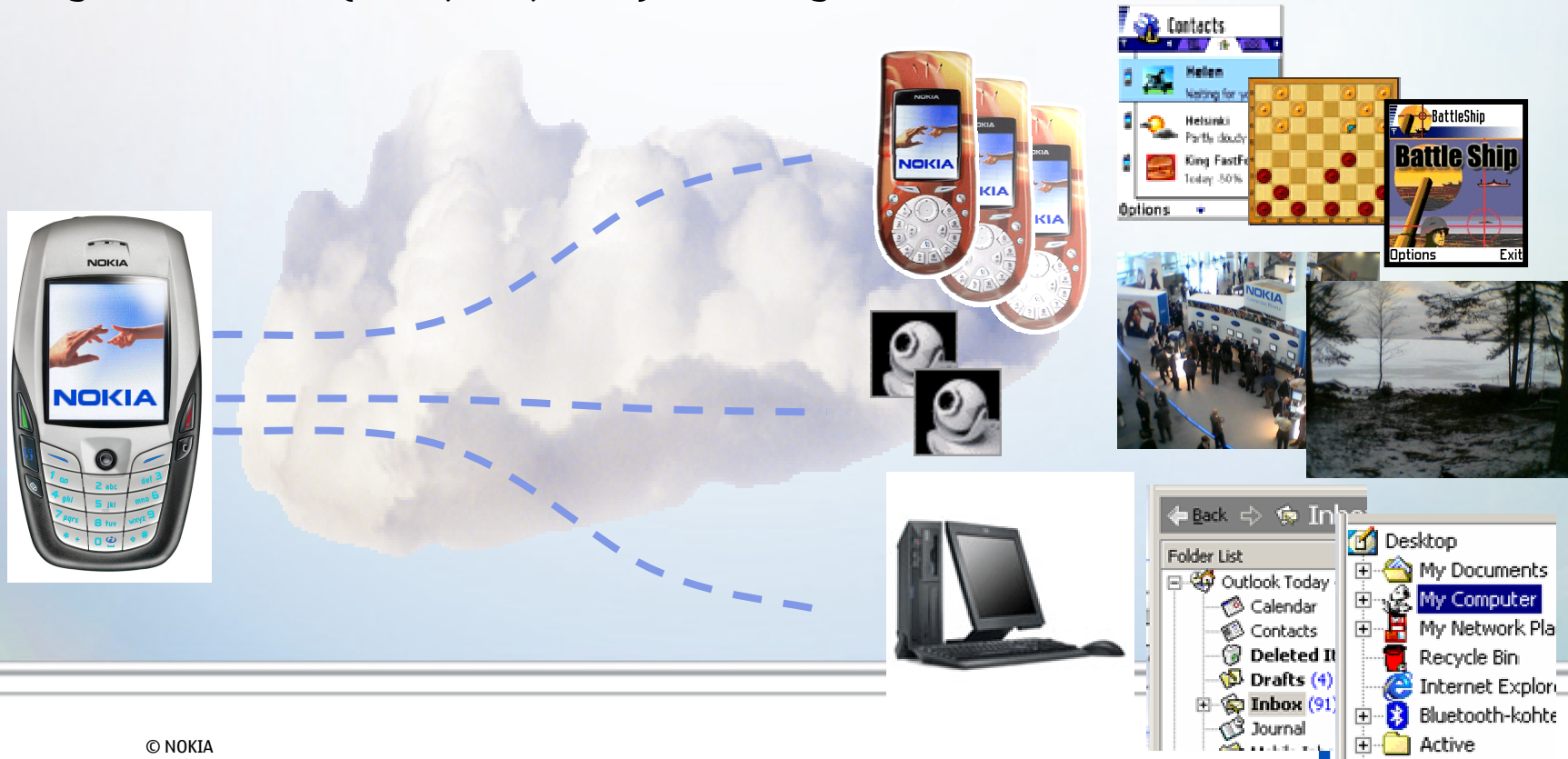


**a direct channel between two terminals
or group of terminals...
...and applications running in the terminals**

Use cases for IP Multimedia

Peer-to-Peer connectivity can be used for various purposes

- Sharing data & media between terminals (gaming, pictures, music, contacts, PoC ...)
- Contacting other devices (PDA's, PC's, cams) and using data which is stored in there



Interactive applications – how

- Standard toolkit for application developers
- Standard SIP stack for mobile terminals
- Standard programming interface (API) for the SIP stack

Nokia SIP plug-in for Series 60 SDK

Nokia SIP plug-in for S60 SDK - enables services

Enables...

- ***Lower threshold*** to develop SIP enabled applications - SIP API hides the network functionality from the developed application
- ***Fast development times*** – industry-leading development tools meaning that developers do not have to be SIP experts
- ***Fast deployment*** – connected services that do not require new servers can be deployed quickly
- ***Leveraging Nokia's mobile penetration*** – developed applications have potential multi-million size market. Starting with S60 (Symbian 7.0s based) and expanding to other Nokia platforms.
- ***Less fragmentation*** – emergence of different implementations of SIP could potentially cause fragmentation of services
- Better interoperability of SIP services

Nokia SIP plug-in for S60 SDK - enables services

How...

- **Piloting of the Nokia SIP Plug-in SDK and selected developer applications with major mobile operators**
- **Network infrastructure to enable these new services has been commercially introduced during 2004.**
- **Services can be used in existing 2.5 networks and all future networks**
- **Both Nokia network and terminal business committed to introducing IMS system**
- **Many operators are currently running pilots to prepare for commercial launch of IMS services**

Tools and Support

- **SIP Plug-in for Series 60 SDK provides tools and documentation for developing and testing SIP enabled Applications**
 - SIP extension for S60 2.0 terminal emulator and S60 2.0 terminal
 - SIP stack and API
 - SIP server emulator
 - Example application
 - User's Guide and Programmer's Guide
- **Tool is available for download from Forum Nokia website (<http://www.forum.nokia.com/main.html>)**
- **Standard Forum Nokia developer support for the tool is available**

Series 60 SIP Development environment

- Nokia Series 60 2.0 SDK for Symbian has to be installed before installing the SIP Plug-in
- The whole environment with two Series 60 terminal emulators and SIP Server emulator can be installed and run on one PC
- Testing in the emulator environment can be done by using SIP Server Emulator to emulate SIP Services of the communication Network.
- Provided SIP Server Emulator implements SIP Proxy and Registrar Server functionality.

